📖 Cycle Count Utility: PRD and FSD Overview for Inventory Management System

Cycle Count Utility — PRD + FSD (As-Is)

Item	Value
System	Cycle Count Utility (`Cycle Count.exe`)
Repository	`CycleCount-main`
Document type	Product Requirements Document + Functional Specification Document, describing current behaviour only
Source of truth	Source code in this repository (every file read, batch by batch)
Technology	VB Shop , Windows Forms, .NET Framework 4.8, x86, Microsoft SQL Server (Finesse database), Infragistics UltraWinGrid v11.1, Excel COM Interop
Labels used	**Assumption** = inferred, not provable from the code. **Gap** = missing or defective logic. **Risk** = behaviour that can cause data or user problems

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0. Repository Scan Results (do this first)

0.1 Batch-by-batch file inventory

Batch	File	Type	In build (.vbproj)?	Present?	Purpose / finding
1 – Solution	`Cycle Count.sln`	VS 2017 solution	n/a	Yes	One project: `Cycle Count.vbproj`
1	`Cycle Count.vbproj`	MSBuild VB project	n/a	Yes	WinExe, .NET 4.8, `OptionStrict Off`, x86. References Infragistics 11.1, Excel Interop 15 (from network share `\\clair\s\Finesse`), System.Speech, PresentationFramework
1	`app.config`	.NET config	Yes	Yes	User settings `myMIAPhaseNo`, `MainFormSize`, `MainFormLocation`,

					`MainFormWindowState`; file log listener; runtime v4.8
1	`.gitmodules`	Git	n/a	Yes	Submodule `ClairUtilities` → `https://git.clairglobal.net/Finesse/ClairUtilitities.git`
1	`.gitignore`	Git	n/a	Yes	Ignores `bin/`, `obj/`, `.vs/`, `*.user`, `ClairUtilities/`
2 – My Project	`My Project\Application.myapp` / `Application.Designer.vb`	Generated	Yes	Yes	Startup form `frmCycleCount`; Windows authentication mode; multi-instance allowed;

					settings saved on exit
2	`My Project\AssemblyInfo.vb`	Generated	Yes	Yes	Title "Cycle Count", v1.0.0.0, © 2010
2	`My Project\Resources.resx` / `Resources.Designer.vb`	Resources	Yes	Yes	5 WAV sounds: `bad_beep`, `good_beep`, `user_input_needed`, `FlagPart`, `continuewarning`
2	`My Project\Settings.settings` / `Settings.Designer.vb`	Generated	Yes	Yes	Same 4 user settings as `app.config`
2	`My Project\app.manifest`	Manifest	Yes	Yes	`asInvoker` (no elevation)
2	`My Project`	Licensing	Yes	Yes	Infragistics UltraGrid licence

	`\licenses.licx`				
3 – Modules	`Modules\CycleCountGeneral.vb`	VB module	Yes	Yes	Sound helpers, `gPartFlagged` / `gPartInRepair` globals, `PartDetail` class, unused helpers
3	`Modules\ExcelExport.vb`	VB class	Yes	Yes	Exports the parts grid to Excel via COM
4 – Forms	`Forms\Cycle Count Utility.vb` (+ `.Designer.vb`, `.resx`)	WinForm	Yes	Yes	**Main application screen** (`frmCycleCount`) – about 2,000 lines of logic
4	`Forms\Cycle Count Check Up.vb` (+ `.Designer.vb`, `.resx`)	WinForm	Yes	Yes	Cycle Count Check Up review screen (`frmCycleCountCheckUp`)

4	`Forms\ConfirmAndContinueDialog.vb` (+ `.designer.vb`, `.resx`)	WinForm	Yes	Yes	Generic warning dialog with "Don't show this warning again"
5 – Shared library	`ClairUtilities\**` (15 compile items, 2 resx, README, 2 PNG, stub shproj/proj items)	VB (submodule)	Yes	**No – MISSING**	Git submodule not checked out. Provides the database helpers, barcode duplicate picker, project picker, speech, process launcher, connection string
6 – Assets	`CheckmarkBox.ico`	Icon	Yes (app icon)	Yes	Application icon
6	`checkmark.ico`	Icon	Yes (Content)	Yes	Copied to output; not referenced by code

6	`Resources\*.wav` (5)	Audio	Yes	Yes	Feedback sounds
6	`Clipboard1.ico`, `ReturnedEquipmentReport.ico`, `ReturnedEquipmentReportColor.ico`, `ReturnedEquipmentReportColorRedBlue.ico`	Icon	**No**	Yes	**Unwanted** – not referenced anywhere; names belong to other Finesse apps
6	`Index.dat` (79 bytes), `Storage.dat` (1,024 bytes, all zeros)	Binary	**No**	Yes	**Unwanted** – not .NET, not referenced. **Assumption:** tool or cache artifacts

0.2 .NET Framework authenticity check

Check	Result
Language of all source files	100% VB Shop (`.vb`), Windows Forms designer code, `.resx` – all standard .NET Framework artifacts

Target framework	`.NETFramework,Version=v4.8` (`app.config`, `.vbproj`)
Non-.NET source code (JS, Python, Java, C++ and so on)	**None found**
Files not part of the build	`Index.dat`, `Storage.dat`, `Clipboard1.ico`, 3 × `ReturnedEquipmentReport*.ico` → **safe to remove** (not referenced)
Build blockers	`ClairUtilities` submodule missing → **project will not compile** as checked in. Infragistics DLL hint path points to a local `Program Files (x86)` install; Excel Interop hint path points to network share `\\clair\s\Finesse`

0.3 Dead or unwanted code inside .NET files (compiles, but unused)

Location	Item	Finding
`CycleCountGeneral.vb`	`SetForegroundWindow`, `ShowWindow` (user32 P/Invoke)	Declared, never called
`CycleCountGeneral.vb`	`EmptyStringToNull`	Never called
`CycleCountGeneral.vb`	`gPartInRepair`	Set, never read
`ExcelExport.vb`	`CreateNewWorkbook`, `BuildFolder`, commented `ProjectDataDump`	Never called

`ExcelExport.vb`	`FilePath` parameter of `CycleCountDataSetToExcel`	Ignored (`c:\temp\myCycleCountWorkbook.xlr` is passed but never used)
`Cycle Count Utility.vb`	`StartDate()`, `EndDate()`	Never called
`Cycle Count Utility.vb`	`_dtCurrency`	Loaded at startup, never used
`Cycle Count Utility.vb`	`LastTransferredPart`, `ErrorNumber` in check-in functions	Populated, never used
`Cycle Count Utility.vb`	Large commented blocks (old Excel export, `tsbTransInq_Click`, warehouse-transfer logic)	Dead
`Cycle Count Utility.Designer.vb`	`tsbExcelPartHistory` ("Export Part History to Excel")	`Visible = False`, **no click handler**
`Cycle Count Utility.vb`	`Imports Infragistics.Shared`	Unused
`Cycle Count Check Up.vb`	`Imports Excel`, `Imports System.Drawing.Color`	Unused
`ConfirmandContinueDialog.designer.vb`	Default button text "Checkout" / "Don't Checkout"	Copied from a check-out application; overwritten at runtime with "Check In" / "Don't Check In"
`.vbproj`	`PresentationFramework` (WPF)	Not used by code in this repo. **Assumption:**

	reference	may be used by `ClairUtilities`

0.4 Master lists

Modules / components

#	Module	Files	Purpose
M1	Application shell	`My Project\*`, `app.config`	Startup, settings, Windows auth mode
M2	Cycle Count Utility (main form)	`Forms\Cycle Count Utility.*`	Browse parts, record counts, scan, check in, MIA transfer, launch tools
M3	Cycle Count Check Up	`Forms\Cycle Count Check Up.*`	Review counts by warehouse, date, and user
M4	Confirm & Continue dialog	`Forms\ConfirmandContinueDialog.*`	Warehouse mismatch confirmation
M5	General helpers	`Modules\CycleCountGeneral.vb`	Sounds, flags, `PartDetail`
M6	Excel export	`Modules\ExcelExport.vb`	Parts grid → Excel
M7	ClairUtilities (external, missing)	submodule	DB extension methods, `FinesseConnectionString`, `SQLQuote`, `SQLDate`, `ReplaceNull`, `Get_AppConfigSetting`, `Get_Location`

			`FromString`, `Get_SizeFromString`, `StartFinesseProcess`, `ExistingWindowBehavior`, `BarcodeHandler` + `frmDuplicatePartPicker`, `ProjectPicker` + `ProjectSearch`, `AnnounceFlaggedPart` (Speech), `SQLJammer`, `HighlightTextbox`

APIs / services

There is **no HTTP/REST/SOAP API**. This is a 2-tier desktop client that calls SQL Server directly. The service contracts are:

Type	Name	Used by
Stored procedure	`Get_Cycle_Count_Utility_Parts_List`	Parts list (F07), Enter Barcode (F15)
Stored procedure	`dbo.Update_Barcode_Receipt_Date`	Cycle Count barcode scan (F14)
Stored procedure	`part_checkin_with_barcode`	Check-in (F14, F16)
Stored procedure	`part_checkin_without_barcode`	Check-in non-barcoded (F16)

Stored procedure	`commit_mass_transfer`	Move to MIA (F17)
Scalar/table functions	`dbo.can_edit_inventory()`, `dbo.can_edit_parts()`, `fn_user_menus()`, `dbo.fn_datetime_strip_time()`, `dbo.today()`, `dbo.fn_IsPartFlagged()`, `dbo.fn_IsBarcodeInRepair()`; built-ins `SUSER_SNAME()`, `GETDATE()`, `@@SPID`	Various
External executables (via `StartFinesseProcess`)	`prjpjinvcal.exe`, `Query Tools.exe`, `utilization.exe`, `Add Inventory.exe`, `Device Maintenance.exe`, `Part Maintenance.exe`, `Project Maintenance.exe`, `TransferInquiryTools.exe`	F19
Desktop COM	Microsoft Excel	F21
Web URL	`SysConfig.PODIO_PUBLIC_ISSUE_TRACKING_URL` opened in default browser	F22
OS	Windows audio (WAV), speech (**Assumption:**	F14, F20, F25

	`AnnounceFlaggedPart` uses System.Speech), clipboard		

Database objects

Object	Kind (as used)	Read	Write	Used by
`warehouses`	table/view	✔		F01, F03, F14, F23
`inStorageLocations`	table	✔		F04
`currency`	table	✔ (unused)		F01
`pjtfrusr`	table (user profile)	✔		F01, F13
`inUnitOfWeight`, `inUnitOfLength`	table	✔		F01, F10
`App_Config`	table/view	✔		F01
`app_config`	table/view		DELETE	F24
`dbo.AppConfig`	table/view		INSERT	F24
`mumenus`	table	✔		F02
`inpart`	table (part master)	✔	UPDATE (verify columns)	F05, F06, F10, F11, F15, F16, F23
`inpartsub`	table (part instances, barcodes, locations)	✔	(through SPs)	F05, F06, F12, F15, F16, F17

`glentities`	table (projects/phases)	✔		F05, F06, F09, F12, F16, F17
`pjproptype`	table	✔		F05, F06, F09
`pjjobbudexp`	table	✔		F12
`partlocationhistory`	table	✔		F09
`WarehouseCycleCounts`	table	✔	INSERT	F09, F13, F23
`mudbmsinfo`	table/view	✔		F13
`dbo.MassTransferStaging_v`	view (staging)		DELETE + INSERT	F17
`dbo.SysConfig`	table	✔		F22

PART A — PRODUCT REQUIREMENTS DOCUMENT (PRD)

A1. Executive Summary

Cycle Count Utility is an internal Windows desktop tool for the Finesse inventory and rental-asset system. Warehouse staff use it to:

1. Find which parts in a warehouse need cycle counting. Parts are ranked by utilization and cycle-count priority from the stored procedure `Get_Cycle_Count_Utility_Parts_List`.
2. See where each part physically is (on-hand lines by barcode, project/phase, warehouse, owner).

3. Record that a part has been counted (`WarehouseCycleCounts`).
4. Scan barcodes during a count. In-stock items get their receipt date refreshed; items found on a project can be checked back into stock.
5. Fix inventory on the spot: check items into the user's warehouse, move items to a personal "MIA" (missing) order, verify dimensions and weight.
6. Jump into other Finesse tools (Availability, Query Tools, Utilization, Add/Remove Inventory, Part/Project Maintenance, Transfer Tools).
7. Review who counted what, and when (Cycle Count Check Up).

A2. System / Module Overview

```mermaid
flowchart LR
  U[Warehouse user<br/>Windows login] --> APP[Cycle Count.exe<br/>.NET 4.8 WinForms]
  APP -->|ADO.NET SqlClient<br/>inline SQL + stored procs| DB[(Finesse SQL Server)]
  APP -->|StartFinesseProcess| EXE[Other Finesse EXEs]
  APP -->|COM Interop| XL[Microsoft Excel]
  APP -->|Process.Start URL| WEB[Podio issue tracker]
  APP --> SND[WAV sounds / speech]
  APP -.compile-time.-> CU[ClairUtilities submodule<br/>MISSING in repo]
```

Screen	Entry	Main areas
Cycle Count Utility (`frmCycleCount`)	App startup (maximized)	Top toolbar; Category list; Part Groups list; Parts grid + dimensions status bar; lower toolbar; Details grid (history); Locations grid + Mark As Counted

Cycle Count Check Up (`frmCycleCountCheckUp`)	"Cycle Count Check Up" button	Warehouse combo, Date Filter toggle; Dates grid → Users grid → Parts counted grid
Warning dialog (`ConfirmandContinueDialog`)	Warehouse mismatch during check-in	Message, Check In / Don't Check In, "Don't show this warning again"

A3. User Personas & Permissions (inferred from code)

Persona	How identified in code	Capabilities
P1 – Cycle Counter (any Finesse user)	Windows login → SQL login (`SUSER_SNAME()`); a `pjtfrusr` row is required (startup reads units from it)	Browse, search, scan, Mark As Counted, check in, MIA move, verify dims/weight, Check Up, Excel, copy, launch unsecured tools
P2 – Inventory Editor	`dbo.can_edit_inventory() = 1` **and** menu right for "Add Inventory" / "Device Maintenance"	Add Inventory / Remove Inventory launchers
P3 – Parts Editor	`dbo.can_edit_parts() = 1` **and** menu right "Part Maintenance"	Part Maintenance launcher
P4 – Utilization user	Menu right "Utilization" in `fn_user_menus`	Utilization toolbar button
P5 – Supervisor / Reviewer (**Assumption**)	Not enforced in code	Uses Check Up to review counts per user and date
P6 – User with no warehouse	`_UserWarehouseCode` empty (no	Blocked from "Check this item into your warehouse"

	`warehouses.is_default = 1` row)	

A4. Functional Scope (feature list)

ID	Feature	Screen	Writes data?
F01	Application startup & session initialisation	Main	No
F02	Menu security trimming	Main	No
F03	Warehouse selection	Main	No
F04	Storage location filter	Main	No
F05	Category (commodity) list	Main	No
F06	Part group list	Main	No
F07	Parts list (cycle-count priority)	Main	No
F08	Free-text part search (debounced)	Main	No
F09	Part details pane: Location History / Part Cycle Count History / User Cycle Count History	Main	No
F10	Part dimensions & weight status bar (unit conversion)	Main	No

F11	Verify dimensions / Verify weight	Main	**Yes** (`inpart`)
F12	Locations grid (on-hand lines)	Main	No (UI only)
F13	Mark As Counted	Main	**Yes** (`WarehouseCycleCounts`)
F14	Cycle Count barcode scan	Main	**Yes** (SPs)
F15	Enter Barcode (locate part by barcode)	Main	No
F16	Check item into your warehouse (barcoded / non-barcoded)	Main	**Yes** (SPs)
F17	Move barcode/part to MIA order	Main	**Yes** (staging + SP)
F18	Reset (choose) MIA order	Main	Local setting
F19	Launch external Finesse tools (8)	Main	No
F20	Copy to clipboard	Main	No
F21	Export parts to Excel	Main	No (creates unsaved workbook)
F22	"LAME!" issue tracker link	Main	No
F23	Cycle Count Check Up	Check Up	No

F24	Window layout persistence	Main	**Yes** (`app_config` / `AppConfig` , local settings)
F25	Audio & voice feedback	All	No

A5. User Journeys

J1 – Planned cycle count Open app → default warehouse preselected → pick Category → pick Part Group → parts appear, ranked → select a part → see on-hand locations → physically count → scan barcodes in "Cycle Count Barcodes" (rows turn green or yellow) → click **Mark As Counted** → confirm.

J2 – Found-on-project item during count Scan a barcode that is on a project (not INSTOCK/REPAIR) → prompt "found on project … check it in?" → Yes → checked into the user's default warehouse INSTOCK → row turns green.

J3 – Locate by barcode Type a barcode in the top barcode box → Enter → category, part group, and part are auto-selected. If the part is in another warehouse, the part is injected into the list.

J4 – Missing item Right-click a location row → "Reset your MIA Order" (pick project) → "Move barcode … to your MIA Order" → confirm or enter qty → transferred.

J5 – Data quality Select part → click "Not Verified" (dimensions or weight) in the status bar → stamped as verified by the current user today.

J6 – Review Click "Cycle Count Check Up" → choose warehouse → pick date → pick user → see the parts that user counted.

A6. Business Objectives & Outcomes

Objective	How the code supports it
Accurate inventory records	Count logging, barcode scan refreshes receipt date, check-in of strays
Prioritised counting effort	SP returns `utilization` and `utildatepriority` ; colour bar on utilization
Accountability	Every count/verification stamped with `SUSER_SNAME()` and time;

	Check Up review
Data quality of part master	Dimension/weight verification stamps
Fast resolution	One-click launch of related Finesse tools with context arguments

A7. Non-Functional Observations

Area	Observation
Security – authentication	Windows authentication (`AuthenticationMode.Windows`); DB identity via `SUSER_SNAME()`. **Assumption:** `FinesseConnectionString` (ClairUtilities) uses integrated security
Security – authorisation	Only 4 toolbar items are permission-trimmed (Tag-based) plus 2 DB permission functions. All write features (Mark As Counted, check-in, MIA transfer, verify) are **not** permission-gated in the client
Security – SQL injection	Most SQL is built by string concatenation using `SQLQuote()` (escaping assumed, not visible). Several queries concatenate **without** `SQLQuote`: `Populate_PartHistory` (`partno = '...'`), `Populate_PartCycleCountHistory`, warehouse code in `@selectedwarehouse`
Security – context menu bypass	Right-click "Utilization" in the parts grid is always visible even when the toolbar Utilization button is hidden

Performance	Parts SP timeout 600 s; filter timer can keep firing every 300 ms (see F08); Check Up loads dates twice on open; Excel export is cell-by-cell for the DataSet build, then a bulk range write
Reliability	Many paths have no `Try/Catch` (MIA transfer, Mark As Counted only partial, Part Info, Enter Barcode) → unhandled exceptions crash or show the .NET dialog
Compliance / audit	Audit is limited to `username` /date columns; no client-side logging other than the default file log listener (not written to explicitly)
Usability	Sounds + colour coding for scanning; maximized windows; layout persisted
Platform	x86, Windows only, requires Excel installed for export, Infragistics licence

A8. Business Rules & Constraints (summary — full table in B9)

- Only parts with on-hand > 0 in the selected warehouse on projects whose property type is utilized or available are offered for Category/Part Group.
- Part group = text before the first comma in the part description.
- INSTOCK and REPAIR are special projects; scanned barcodes there are "confirmed" rather than checked in.
- Counts are inserted (history), never updated.
- Checked-in items go to the user's default warehouse, project `INSTOCK`.

A9. Assumptions & Limitations

#	Type	Statement
L1	Gap	`ClairUtilities` source is missing; behaviour of helpers

		(`SQLQuote` , `BarcodeHandler` , `ProjectPicker` , `StartFinesseProcess` , `AnnounceFlaggedPart` , DB extension methods) is inferred from usage
L2	Assumption	`warehouses.is_default` , `App_Config` , `mudbmsinfo` are user-scoped (filtered to the current login) views. Behaviour of F01, F13, F14 depends on this
L3	Assumption	`MassTransferStaging_v` is filtered by session (`@@SPID`), so `delete dbo.MassTransferStaging_v` only clears the current session's rows
L4	Limitation	No offline mode; every action hits SQL Server synchronously on the UI thread
L5	Limitation	Single DataGridView selection; one part at a time for Mark As Counted
L6	Limitation	Business logic for priority, check-in, transfer lives in stored

		procedures not in this repo

PART B — FUNCTIONAL SPECIFICATION DOCUMENT (FSD)

Line references are in the form `Utility.vb:123` (`Forms\Cycle Count Utility.vb`), `CheckUp.vb:…`, `Designer:…`.

B1. Feature-by-Feature Functional Specs

F01 — Application Startup & Session Initialisation

Purpose: Prepare the session, permissions, reference data, and window layout. (`Utility.vb:203-274`)

Aspect	Detail
Trigger	`frmCycleCount_Load` (startup form)
Inputs	Windows login; `My.Settings`; DB
Processing	1. Set split panel minimum sizes. 2. Open the shared connection `gconn` (`FinesseConnectionString`). 3. Run one batch returning 7 result sets: `can_edit_inventory()`, `can_edit_parts()`, `currency`, `warehouses` (excluding `LAX2`, ordered by description), storage locations (`<Storage Location>` + distinct `inStorageLocations`), current user profile from `pjtfrusr` joined to unit tables where `user_name = suser_sname()`, all of

	`dbo.App_Config`. 4. `SecureMenu()` (F02). 5. Fill warehouses (F03) and storage locations (F04). 6. MIA phase = `App_Config["myMIAPhaseNo"]`, else `My.Settings.myMIAPhaseNo`; saved back to `My.Settings`. If not empty, read `glentities.entitydesc` for the MIA description and rename the menu items. 7. Restore window state: Maximized → location then maximize; Normal → location + size (DB value first, then local setting). Minimized is not handled. 8. `IsDataLoaded = True`
Outputs	Populated combos; permission-trimmed toolbar
Errors	No `Try/Catch`. If the user has no `pjtfrusr` row (or no unit mapping), `_dtUserInfo.Rows(0)` fails later in F10 (**Risk**)

F02 — Menu Security Trimming (`Utility.vb:276-354`)

Aspect	Detail
Inputs	`Tag` values on `ToolStrip1` items (recursive into drop-downs): `Utilization`, `Add Inventory`, `Device Maintenance`, `Part Maintenance`
Processing	Query `fn_user_menus(suser_sname())` ⋈ `mumenus` where `appframe IN (tags)` and

	`effective_excluded = 0`. Each tagged item `.Available = (tag in allowed list)`. If `can_edit_inventory = 0` → hide Add Inventory and Remove Inventory. If `can_edit_parts = 0` → hide Part Maintenance. Parent drop-downs (Adjust Inventory) are hidden when no child is available (deepest parent first)
Outputs	Toolbar visibility
Gaps	Items without a Tag (Availability, Query Tools, Refresh, Excel, Enter Barcode, LAME, all lower toolbar items, context menus, Mark As Counted) are never secured. Context menu "Utilization" is not secured

F03 — Warehouse Selection (`Utility.vb:356-373, 880-888, 51-53`)

Aspect	Detail
Inputs	`warehouses` rows; display format `"<code> - <description>"`
Processing	Fill combo; select the first row with `is_default = True`; store `_UserWarehouseCode` and `myWarehouseDescription`. On change: clear Category, Part Groups, grids; reload categories (F05). `CurrentWarehouseCode()` = text before `" -"`
Output	Combo is resized on Paint to fit the text (+50 px)
Edge	If no default warehouse, nothing is selected and

	`_UserWarehouseCode` stays empty → F16 blocked; F05 would fail on `CurrentWarehouseCode` (**Risk**)

F04 — Storage Location Filter (`Utility.vb:375-382, 1299-1312`)

Aspect	Detail
Inputs	`tscboStorageLocation` (editable combo; first item `<Storage Location>`)
Processing	On selection change: if not `<Storage Location>` , `bsParts.Filter = [strFilter AND] storagelocation = '<value>'` ; else `bsParts.Filter = strFilter`
Output	Client-side filter on the loaded parts list
Edge	The combo is editable (DropDown); typed text is not applied until an item is selected

F05 — Category (Commodity) List (`Utility.vb:385-423`)

Aspect	Detail
Processing	`ALL` + distinct `inpart.commmodity` where not null/''/'UNUSED', joined to `inpartsub` (`bld` = current warehouse, `onhand > 0`) → `glentities` → `pjproptype` (`isutilized = 1 OR isAvailable = 1`)
On select	Wait cursor, clear Part Groups and grids, load Part Groups (F06)

F06 — Part Group List (`Utility.vb:425-458, 1156-1158`)

Aspect	Detail
Processing	`ALL` + distinct `partdesc_cat` = text before the first comma of `inpart.partdesc` (descriptions without a comma are excluded), same warehouse/on-hand/property-type filters, `commmodity <> 'UNUSED'`, plus the selected commodity unless `ALL`
On select	`Populate_Parts()` (F07)

F07 — Parts List (`Utility.vb:460-503, 1983-2011, 529-560`)

Aspect	Detail
Preconditions	`IsDataLoaded`; a Part Group is selected, else message **"No Part Group has been selected"**
Call	SP `Get_Cycle_Count_Utility_Parts_List` on a new connection, `CommandTimeout = 600`: `@warehouse` always; `@commodity` if not `ALL`; `@partgroup` if not `ALL`
Columns (from SP)	`partno` "Part No", `t_qty` "WH Total Qty", `a_qty` "WH Avail Qty", `partdesc` "Part Description", `utilization` "% Utilized" (format `#0%`), `lastcyclecount` "Last Cycle Count", `utildatepriority` "Cycle Count Priority" (`#0%`), `empname` (hidden, "Counted by"), `username`

	(hidden), `storagelocation` (hidden)
Rendering	Column index 4 (**Assumption:** `utilization`) drawn as a bar: value 0 → whole cell red; fill width = value × cell width; < 20% light pink; ≥ 80% light green; otherwise powder blue
Post	Load details (F09/F10); enable **Mark As Counted** only if rows > 0
Grid	Read-only, full-row select, single select, no add/delete/resize

F08 — Free-text Part Search (`Utility.vb:1164-1235, 1457-1459`, Designer timer 300 ms)

Aspect	Detail
Trigger	Typing in `tstxtFilter` restarts a 300 ms timer; on tick the filter runs
Processing	1. Remember the current part. 2. If the previous search used the "all groups" fallback, reload the parts. 3. Split text on spaces; build `partno like '%' AND (partno like '%x%' OR partdesc like '%x%')` for each word (AND across words). 4. Same as last filter → exit. 5. Apply via storage-location logic (F04). 6. If the current part changed → reload details and locations. 7. If no rows match and the fallback has not run → `Populate_Parts(ExcludePartGroupFilters:=True)`. 8. Stop the timer
Gap	**Fallback ineffective:** `Populate_Parts` calls

	`Get_Parts_From_Warehouse(CurrentWarehouseCode)` without passing the flag, which resets `PartGroupFiltersExcluded = False` and still sends the commodity/part-group filters
Gap	**Timer never stops when the search is unchanged:** the early `Exit Sub` skips `Timer_tstxtFilter.Stop()`, so the timer keeps firing every 300 ms
Edge	DataView filter syntax: characters such as `[`, `]`, `*` in the search text may raise a filter exception (escaping by `SQLQuote` assumed only for quotes)

F09 — Part Details Pane (`Utility.vb:573-682, 742-769, 1059-1094`)

Selecting a part (BindingSource `CurrentChanged`) sets `txtSpUserName` = part's `username` (enables "Selected User Cycle Count History" only if not empty), loads details, shows **Last Cycle Count** (`lastcyclecount`) and **by:** (`empname`), clears scan highlights, and loads Locations (F12). The **Display** drop-down chooses one mode (radio-like; label `tslSubDetail` shows the mode):

Mode	Query	Columns
Part Location History (default)	`partlocationhistory` ⋈ `glentities` ⋈ `pjproptype` for the last 1 year (`startdate < now` and `isnull(enddate, now) >= now-1y`), `partno`, `bld` = warehouse, `g.industry` or `g.agency` in	Phase No, Qty, Phase Description, Start Date, End Date (null → `dbo.Today()`); End Date = today highlighted **yellow**

	(warehouse, ''), `t.isavailable = 1`, ordered by start desc, end desc	
Part Cycle Count History	`warehousecyclecounts` where `partno` and `username is not null` (all warehouses)	Warehouse, Date Counted, Counted By
My Cycle Count History	`warehousecyclecounts` ⋈ `inpart` where `username = suser_sname()` ordered by date desc	Part No, Part Desc, Date Counted
Selected User Cycle Count History	same, `username = <part's last counter>`	same

F10 — Part Dimensions & Weight Status Bar (`Utility.vb:1096-1154`)

Aspect	Detail
Inputs	`inpart.length_uom`, `width_uom`, `depth_uom` (inches), `projpercent` (used as **weight in lbs**), verification columns; user's `conversionFactorFromInches`, `conversionFactorFromPounds`, unit short names
Processing	Null → 0; value × factor rounded to 2 decimals + unit. Verified date present → "Verified `<date> by <user>" in green, else "Not Verified" in red`
Edge	Part not found → `Rows(0)` exception (**Risk**). Wait cursor is not reset on the early `Return` path

F11 — Verify Dimensions / Weight (`Utility.vb:1461-1495`)

Aspect	Detail
Trigger	Click the dims or weight verified label
Preconditions	`IsDataLoaded`, parts exist, a selection exists
DB write	`UPDATE inpart SET dimsVerifiedDate = dbo.today(), dimsVerifiedUser = suser_sname()` (or `weightVerified*`) `WHERE partno = <selected>`
Notes	No confirmation; re-click re-stamps; not gated by `can_edit_parts`

F12 — Locations Grid (`Utility.vb:771-788, 1314-1329, 1440-1455`)

Aspect	Detail
Query	`inpartsub` ⋈ `glentities` ⋈ `pjjobbudexp` (entity + part) where `partno`, `bld` = current warehouse, `onhand > 0`, ordered by `batchno` desc, `unique_no` asc
Columns	Qty, Barcode, Phase No, Phase Desc, WH, Owner (`dept`), End Date (`todate`), Receipt, Parent BC
Row formatting	Scanned barcode (in `aryHighlightedBarcodes`) → lime green background; `todate < today` → red text; has parent barcode → grey italic

Edge	`CellClickAction = Edit` : cells can be edited in the UI but are never saved

F13 — Mark As Counted (`Utility.vb:927-957`)

Aspect	Detail
Enabled	Only when the parts grid has rows (F07)
Validation	No selected cell or empty value → **"A part must be selected to Mark as Cycle Counted."**
Confirmation	**"Are you sure you want to mark this item as cycle counted for YOUR warehouse?"** (Yes/No/Cancel; only Yes proceeds)
DB write	`INSERT INTO WarehouseCycleCounts (warehouse, partno, lastcount, username) SELECT warehouse_entity, '<partno>', GETDATE(), SUSER_SNAME() FROM pjtfrusr p JOIN mudbmsinfo i ON p.user_name = i.username`
Scope	Only the selected part (top grid). Locations-grid rows, barcodes, quantities, and the warehouse combo are **not** used
Post	Reload parts (F07)
Errors	`Try/Catch` → `MsgBox(ex.Message)`
Risk	The SELECT has no `suser_sname()` filter. If `mudbmsinfo` is **not** a current-user

	view, one row is inserted per user in the join (each with that user's warehouse). **Assumption:** `mudbmsinfo` is scoped to the current session, giving one row in the user's own warehouse

F14 — Cycle Count Barcode Scan (`Utility.vb:96-175`)

```mermaid
flowchart TD
  A[Enter pressed, Cycle Count Barcodes box not empty] --> B[BarcodeHandler duplicate check]
  B -->|Cancelled| X[Stop]
  B -->|Duplicate chosen / different barcode| C[Use picked barcode]
  B -->|No dup| C
  C --> D[Clear textbox]
  D --> E{Barcode in Locations grid<br/>of current part?}
  E -->|No| F[Beep_Fail]
  E -->|Yes| G{Phase = INSTOCK or REPAIR?}
  G -->|Yes| H[exec Update_Barcode_Receipt_Date<br/>@barcode, OUT @IsFlagged]
  H -->|Flagged| I[Fail beep + announce, row YELLOW]
  H -->|Not flagged| J[Success beep + announce, row GREEN]
  G -->|No| K[Prompt: found on project X - check it in?]
  K -->|Yes| L[CheckInWithBarcode to default WH, INSTOCK<br/>update row cells locally, row GREEN]
  K -->|No| M[No action]
  I & J & L & M --> N[Add barcode to highlight list]
```

Aspect	Detail

Inputs	`tstxtCycleCountBarcodes` + Enter (form `KeyPreview = True`)
Duplicate handling	`BarcodeHandler(barcode, Me).ShowPickerDialogGeneral(True, "", True)` (ClairUtilities); may replace the barcode with the chosen `InpartSubRecord.UniqueNo`; `_DuplicateBarcode` flag passed to `AnnounceFlaggedPart`
Scope	Matches **only** rows currently loaded for the selected part and warehouse
Check-in path	Warehouse = `select warehouse_code from warehouses where is_default = 1`; calls `CheckInWithBarcode(barcode, batchno, userWH, "INSTOCK")`; locally sets `receipt = Now`, `batchno = entitydesc = "INSTOCK"`, `bld = userWH`
Gaps	The return value of `CheckInWithBarcode` is ignored: the row turns green and the local cells change even if check-in failed. No warehouse-mismatch confirmation (unlike F16). Prompt answer "No" still adds the barcode to the highlight list (green on next refresh)

F15 — Enter Barcode (Locate Part) (`Utility.vb:1905-1981, 2013-2017`)

Aspect	Detail

Inputs	`tstxtEnteredBC` (trimmed) + Enter key or the Enter Barcode button; empty → ignored
Processing	Duplicate handling (as F14). Query `inpartsub` ⋈ `inpart` for `unique_no` where `onhand > 0 OR parentunique_no <> ''` → `partno`, `bld`, `commmodity`, `partdesc_cat`. Not found → **"Barcode does not exist"**. Else select the commodity (add it to the list if missing), select the part group (add if missing), find the part in `bsParts`. If absent, run the parts SP for the **barcode's** warehouse; copy the part row with `t_qty`, `a_qty`, `utilization` = NULL, `lastcyclecount = 1900-01-01`, `utildatepriority = 0`, then add and select it
Edge	Null `commmodity` or no comma in the description (`partdesc_cat` NULL) → cast exception (**Risk**). The warehouse combo is not switched to the barcode's warehouse, so the locations grid still shows the current warehouse

F16 — Check Item Into Your Warehouse (`Utility.vb:1497-1878`)

Context menu on the Locations grid. Visible only when the active row's phase is not `INSTOCK` / `REPAIR`. Text: *"Check <barcode X | part Y> into your warehouse: `<desc> INSTOCK"*.

```mermaid
sequenceDiagram
  actor U as User
  participant F as frmCycleCount
```

```
  participant DB as Finesse SQL
  U->>F: Right-click row → Check into warehouse
  F->>F: Validate row selected, user warehouse set
  alt Barcoded
    F->>U: Confirm (Yes/No/Cancel)
    F->>DB: glentities/inpartsub warehouse lookups
    opt WH mismatch and not suppressed
      F->>U: ConfirmandContinueDialog (Check In / Don't)
    end
    F->>DB: fn_IsPartFlagged / fn_IsBarcodeInRepair
    F->>DB: exec part_checkin_with_barcode
    DB-->>F: ErrorMessage, ErrorType, Return*, IsFlagged, RepairTicketExists
  else Non-barcoded
    F->>U: InputBox qty (default 0)
    F->>DB: exec part_checkin_without_barcode
    DB-->>F: ErrorMessage, ErrorType, ReturnPartDesc, IsFlagged
  end
  F->>F: Sounds / error message
  F->>DB: Reload locations
```

Validations

Rule	Message
No active location row	"Nothing has been selected to be checked in."
`_UserWarehouseCode` empty	"You do not belong to a warehouse. Contact your IT adminstrator for assistance."

Barcoded confirm	"Check the following barcoded item into `<WH> INSTOCK? … Barcode: … <part>: <desc>" (only Yes proceeds)`
Non-barcoded qty not numeric (incl. Cancel/blank)	"`<value> is not a number. Operation aborted." / "Operation aborted."`
Qty > on-hand	"The quantity of `<n> you entered to be checked in is greater than the quantity available to be checked in (<onhand>). Operation aborted."`
Qty ≤ 0	Silently does nothing (locations still reloaded)

Warehouse confirmation (ProjectOrBarcodeWarehouseConfirmation)

- No phase → OK.
- Project warehouse = `isnull(nullif(industry,''), agency)`; barcode warehouse = `inpartsub.bld` if the barcode is on INSTOCK (because the INSTOCK agency is "LITZ"), else the project warehouse of the barcode's entity.
- Phase INSTOCK/REPAIR → OK. Suppressed for the session ("Don't show again"), invalid barcode, barcode equals a project number, or warehouses match → OK.
- Otherwise beep and show *"The selected warehouse, {0} does not match the project warehouse, {1}. Do you want to continue?"*.
- A `NullReferenceException` while reading results → treated as OK (explicit hack in code).

part_checkin_with_barcode result handling

- No error **and** the returned batch is a prefix-parent of the phase (e.g. returned `123-BM`, phase `123-BM-6`) → treated as `CheckedOutToOtherProject` → warning beep, return False.
- No error otherwise → success.
- `BarcodeIsProjectNumber` / `CheckedOutToOtherProject` → warning beep, no message.
- Any other error (or `SqlException`) → fail beep, **form background turns red (never reset)**, `MsgBox(ErrorMessage, title "<Phase> Error")`.

part_checkin_without_barcode – success = empty `@ErrorMessage` → success beep. `PartNotIncluded`, `PartialCheckIn` → fail beep, no message (commented logic). Other → message box.

Sounds: barcoded success path plays the check-in sound **twice** (before and after the SP); the flagged sound plays if `fn_IsPartFlagged` / `@IsFlagged`.

Risks: `CheckInBarcodedPart` reads `t.Rows(0)` with `onhand = 1`. A child or zero-on-hand barcode causes an `IndexOutOfRange` crash. No selected part (`dvSelectedRow Nothing`) → `NullReference`. DBNull output parameters cast to String → `InvalidCastException` (only `SqlException` is caught).

F17 — Move Barcode/Part to MIA Order (`Utility.vb:1331-1410, 1428-1438, 177-201`)

Aspect	Detail
Visibility	Shown when `My.Settings.myMIAPhaseNo` is not empty. Text: "Move <barcode X / part Y> to your MIA Order: `<phase> (<desc>)"`
Validations	No part → "No part has been selected."; no row → "Nothing has been selected to move to MIA."; no MIA → "No MIA Order has been selected."
Barcoded	Qty = 1; confirm "Are you sure you want to transfer barcode X to MIA order P: D?"
Non-barcoded	InputBox "How many of (`<onhand>) <part>: <desc> (<WH>, <owner>) would you like to move from <phase>: <desc> to <MIA>: <desc>?" default "0". Blank → cancel. Non-numeric or > on-hand → "Invalid qty entered."`
DB	MIA warehouse = `glentities.agency` of the MIA

	phase. Batch: `DELETE dbo.MassTransferStaging_v; INSERT … SELECT @@spid, qty, partno, unique_no, serial_no, batchno→MIA, dept→dept, area→area, bld→MIA WH, row, parentunique_no FROM inpartsub WHERE partno, unique_no, batchno, onhand > 0, bld, dept, parentunique_no = ''; EXEC commit_mass_transfer`
Post	Success beep (unconditional); reload locations
Gaps	No `Try/Catch` (SQL errors are unhandled). Negative qty passes validation (`CInt(-5) > onhand` is False). Decimal qty passes `IsNumeric` but `Integer.Parse("1.5")` throws. Child barcodes (`parentunique_no <> ''`) insert nothing, yet the success beep still plays. The `ResetMIAOrder` visibility line is overwritten logic (menu always rebuilt)

F18 — Reset (Select) MIA Order (`Utility.vb:1412-1426`)

`ProjectPicker(Me, mnuLocationsActions).GetProject()` (ClairUtilities; static instance). On OK: `My.Settings.myMIAPhaseNo = ProjectNum`, store the description, rename menu items. Persisted locally only (saved on exit); **not** written to `App_Config` (startup prefers the DB value if one exists → **Gap**: local choice can be overridden).

F19 — Launch External Finesse Tools

Button / menu	Precondition	Executable	Arguments

Availability (toolbar + parts context)	Parts rows > 0, else "No part has been selected."	`prjpjinvcal.exe`	`partno=<part>`
Query Tools	same	`Query Tools.exe` (BringToFront if running)	`bld:<WH> partno:<part>`
Utilization (secured toolbar; context unsecured)	same	`utilization.exe`	`bld=<WH> partno=<part>`
Adjust Inventory → Add Inventory	secured + `can_edit_inventory`	`Add Inventory.exe`	`serialno= partno=<part>`
Adjust Inventory → Remove Inventory	secured (tag "Device Maintenance") + `can_edit_inventory`	`Device Maintenance.exe`	`partno=<part>`
Adjust Inventory → Part Maintenance	secured + `can_edit_parts`	`Part Maintenance.exe`	`partno=<part>` (no row check → **Risk**)
Project Maintenance (lower toolbar + details context)	none	`Project Maintenance.exe`	`entityno` of the current details row; blank if INSTOCK or no rows
Transfer Tools	none	`TransferInquiryTools.exe`	`entityno:<entity>`; blank if INSTOCK

Context menu `mnuDataGrid` shows part items when right-clicking or clicking the parts grid, and project items for the details grid. Project/Transfer read `DataBoundItem("entityno")`, which exists only in Location History mode (other modes → exception, **Risk**).

F20 — Copy to Clipboard (`Utility.vb:684-740, 1283-1297`)

Item	Copies
Copy Part Number / Description	Selected parts row `partno` / `partdesc`
Copy ALL Data (parts or details)	All rows: header row once, then rows (tab-separated via grid clipboard). Clipboard failure → "The Clipboard could not be accessed."
Copy Phase Number / Description	Details grid `SelectedCells(0)` / `SelectedCells(2)` (**Risk:** relies on the selected-cell order)

F21 — Export Parts to Excel (`Utility.vb:801-878`, `ExcelExport.vb`)

Aspect	Detail
Validation	No columns or rows → "No data is available for export." (title "Export to Excel")
Processing	Copy every grid column (by header text, including hidden columns) and row into DataTable "Parts". Exclude the column named `username`. New Excel instance (hidden), sheet "Cycle Count", bulk write, left-aligned, columns E and G formatted `#0%`, header bold, autofit A:AY, then make Excel visible. The workbook is **not saved**
Notes	Hidden columns "Counted by" and "Storage Location" **are** exported. Columns E/G formatting assumes fixed positions (**Assumption**).

	Column-letter math has an operator-precedence defect for more than 26 columns. Excel must be installed

F22 — "LAME!" Link (`Utility.vb:1888-1903`)

Reads `SysConfig.value` where `tag = 'PODIO_PUBLIC_ISSUE_TRACKING_URL'` and calls `Process.Start(url)`. A null or empty value → unhandled exception.

F23 — Cycle Count Check Up (`CheckUp.vb`, entire file)

```mermaid
flowchart LR
  W[Warehouse combo<br/>excl. NEW, LAX2; default selected] --> D[Dates grid:<br/>date, parts counted]
  D -->|select date| U[Users grid:<br/>username, parts counted]
  U -->|select user| P[Parts grid:<br/>Part No, Desc, Last Cycle Counted]
  T[Date Filter toggle] -->|off| U2[Hide dates; users & parts across all dates]
```

Query	SQL logic
Dates	`warehousecyclecounts` where `username is not null`, `warehouse = WH`, grouped by `fn_datetime_strip_time(lastcount)`, desc
Users	grouped by `username` (+ date if Date Filter is on), `count(username)`
Parts	`warehousecyclecounts` ⋈ `inpart` for user + WH (+ date), ordered by `lastcount` desc

Notes: own connection opened at Load and never closed explicitly; dates are queried twice on open (combo change + Load); the parts grid is editable in the UI (not saved); no permission check. The default form instance is reused (`frmCycleCountCheckUp.Show()`).

F24 — Window Layout Persistence (`Utility.vb:59-94`)

On closing: save location, size, and state to `My.Settings`, then for each setting run `DELETE FROM app_config WHERE cmndsel = 'CYCLECOUNT' AND tag = …` and `INSERT dbo.AppConfig(cmndsel, tag, value)`. All exceptions are swallowed. **Gap:** delete targets `app_config`, insert targets `AppConfig`, and startup reads `App_Config` – three different names. **Assumption:** these are synonyms or views; otherwise rows accumulate or are never read.

F25 — Audio & Voice Feedback (`CycleCountGeneral.vb`)

Event	Sound
Success	`good_beep.wav`
Failure / not found	`bad_beep.wav`
Needs user input	`user_input_needed.wav`
Warning (`BarcodeIsProjectNumber`, `CheckedOutToOtherProject`)	`continuewarning.wav`
Flagged part (check-in)	`FlagPart.wav` (then resets `gPartFlagged`)
Scan announcement	`AnnounceFlaggedPart(barcode, isDuplicate)` – ClairUtilities (**Assumption:** text-to-speech)

B2. Input Rules + Validation Logic (Validation Rules Table)

VR	Field / action	Rule	Behaviour on failure	Source

VR01	Part Group selection	Must be selected before loading parts	"No Part Group has been selected"	`Utility.vb:466`
VR02	Mark As Counted	A part cell must be selected and not empty	"A part must be selected to Mark as Cycle Counted."	`:929`
VR03	Mark As Counted	Explicit Yes confirmation	Abort	`:934`
VR04	Mark As Counted button	Enabled only when parts rows > 0	Disabled	`:501`
VR05	Cycle Count barcode	Textbox not empty, Enter key	Ignored	`:97`
VR06	Cycle Count barcode	Must exist in the current Locations grid	Fail beep, no message	`:172`
VR07	Enter Barcode	Trimmed, not empty	Ignored	`:1907`
VR08	Enter Barcode	Must exist with on-hand > 0 or a parent	"Barcode does not exist"	`:1943`
VR09	Check-in	Location row selected	"Nothing has been selected to be checked in."	`:1505`
VR10	Check-in	User has a default warehouse	"You do not belong to a	`:1510`

			warehouse. ..."	
VR11	Check-in non-barcoded qty	Numeric	"`<x> is not a number. Operation aborted."`	`:1537`
VR12	Check-in non-barcoded qty	≤ on-hand	"The quantity of ... Operation aborted."	`:1532`
VR13	Check-in non-barcoded qty	> 0	Silently skipped	`:1528`
VR14	Check-in barcoded	Warehouse must match the project/barcode warehouse, or the user confirms	Confirm dialog	`:1849-1863`
VR15	MIA move	Part selected	"No part has been selected."	`:1337`
VR16	MIA move	Location row selected	"Nothing has been selected to move to MIA."	`:1344`
VR17	MIA move	MIA order configured	"No MIA Order has been selected."	`:1349`
VR18	MIA move qty	Not blank; numeric; ≤	Cancel / "Invalid qty	`:1362-1368`

		on-hand	entered."	
VR19	Launch tools (Availability, Query, Utilization, Add/Remove Inventory)	Parts rows > 0	"No part has been selected."	`:960…1036`
VR20	Excel export	Grid has columns and rows	"No data is available for export."	`:802`
VR21	Verify dims/weight, Part info	Data loaded, rows and selection exist	Silently ignored	`:1462, 1097`
VR22	Selected-user history menu	Enabled only if the part has a last counter	Disabled	`:755`
VR23	Check-in menu visibility	Hidden for INSTOCK / REPAIR rows	Hidden	`:198`
VR24	MIA menu visibility	Hidden when no MIA phase	Hidden	`:1432`

B3. Processing Logic (step-by-step)

Step-by-step logic is given for each feature in B1 (F01–F25). Cross-cutting processing rules:

1. All SQL runs synchronously on the UI thread; a wait cursor is shown in most long operations.
2. The shared connection `gconn` stays open for the form's lifetime; SPs and writes often open short-lived connections.
3. Data binding: parts → `BindingSource bsParts` → `dgParts`; filters are client-side (`BindingSource.Filter`).
4. `IsDataLoaded` acts as a re-entrancy guard during loads and filtering.

B4. Output + DB Operations + "API" (Stored Procedure) Contracts

B4.1 Write operations

#	Feature	Statement	Table	Values
W1	F13	INSERT… SELECT	`Warehouse CycleCounts`	warehouse = `pjtfrusr.warehouse_entity`, partno, `GETDATE()`, `SUSER_SNAME()`
W2	F11	UPDATE	`inpart`	`dimsVerifiedDate/User` or `weightVerifiedDate/User` = `dbo.today()`, `suser_sname()`
W3	F14	EXEC	`Update_Barcode_Receipt_Date`	**Assumption:** updates the barcode receipt date
W4	F14/F16	EXEC	`part_checkin_with_barcode`	see contract
W5	F16	EXEC	`part_checkin_without_barcode`	see contract
W6	F17	DELETE + INSERT + EXEC	`MassTransferStaging_v`, `commit_ma`	see F17

			`ss_transfer`	
W7	F24	DELETE + INSERT	`app_config` / `AppConfig`	`cmndsel = 'CYCLECOUNT'`, tag, value

B4.2 Stored procedure contracts

There are no HTTP headers or status codes. The equivalents are input/output parameters and SQL exceptions.

Get_Cycle_Count_Utility_Parts_List (CommandType StoredProcedure, timeout 600 s)

Param	Dir	Type	Required	Notes
`@warehouse`	IN	string	Yes	Warehouse code
`@commodity`	IN	string	No	Omitted when `ALL`
`@partgroup`	IN	string	No	Omitted when `ALL`

Result set	OUT	rows		`partno`, `t_qty`, `a_qty`, `partdesc`, `utilization`, `lastcyclecount`, `utildatepriority`, `empname`, `username`, `storagelocation` (order assumed; column 4 = utilization)

dbo.Update_Barcode_Receipt_Date

Param	Dir	Type
`@barcode`	IN	varchar(50)
`@IsFlagged`	OUT	bit

part_checkin_with_barcode (parameters derived at runtime via `SqlCommandBuilder.DeriveParameters`)

Param	Dir	Value sent
`@unique_no`	IN	barcode
`@from_batchno`	IN	current phase
`@parent_batchno`	IN	current phase
`@to_bld`	IN	user warehouse
`@to_batchno`	IN	`INSTOCK`

`@hold_in_warehouse`	IN	False
`@allow_warehouse_transfer`	IN	True
`@ErrorMessage`, `@ErrorType`	OUT	error info (`BarcodeIsProjectNumber`, `CheckedOutToOtherProject`, others)
`@ReturnPartNo`, `@ReturnBatchNo`, `@ReturnPartDesc`	OUT	resulting part/phase
`@IsFlagged`, `@RepairTicketExists`	OUT	bit flags

part_checkin_without_barcode

Param	Dir	Value
`@partno`	IN	part
`@qty`	IN	integer qty
`@from_batchno`	IN	phase
`@to_bld`	IN	user warehouse
`@to_batchno`	IN	`INSTOCK`
`@ErrorMessage`, `@ErrorType` (`PartNotIncluded`, `PartialCheckIn`, …), `@ReturnPartDesc`, `@IsFlagged`	OUT	

commit_mass_transfer – no parameters; processes the session's rows in `MassTransferStaging_v` (**Assumption**).

Functions used: `dbo.can_edit_inventory()` → bit; `dbo.can_edit_parts()` → bit; `fn_user_menus(user)` → (`packagecd` , `menuname` , `menuseqno` , `effective_excluded`); `dbo.fn_IsPartFlagged(partno, toBatchNo, unique_no, wh)` → bit; `dbo.fn_IsBarcodeInRepair(partno, unique_no)` → bit; `dbo.fn_datetime_strip_time(dt)` ; `dbo.today()` .

B5. Flow Diagrams (Napkin AI - Turn Text into AI Diagrams and Visuals prompts + Mermaid)

Paste each prompt into Napkin AI - Turn Text into AI Diagrams and Visuals ("Generate visual") to produce a diagram. Mermaid versions for F14, F16, and F23 are embedded above.

Flow	Napkin AI - Turn Text into AI Diagrams and Visuals prompt
Startup	"Flow: User opens Cycle Count Utility → connect to Finesse SQL → load permissions (can edit inventory, can edit parts), warehouses, storage locations, user units, app config → trim toolbar by menu rights → preselect default warehouse → load categories → restore window layout."
Browse & count	"Flow: Select warehouse → categories → part groups → stored procedure returns parts ranked by utilization and priority → select part → show location history, dimensions, on-hand locations → click Mark As Counted → confirm → insert WarehouseCycleCounts row with user and time → refresh list."
Barcode scan	"Decision flow: scan barcode → duplicate check → is barcode in current part's locations? no → fail beep. yes → is phase INSTOCK or REPAIR? yes → update receipt date → flagged? yellow : green. no → ask to check in → yes → check in to

	default warehouse INSTOCK → green."
Check-in	"Sequence: user right-clicks location → validations → barcoded: confirm, warehouse mismatch dialog, flag check, part_checkin_with_barcode; non-barcoded: enter qty, part_checkin_without_barcode → sounds/errors → refresh locations."
MIA transfer	"Flow: choose MIA order via project picker → right-click location → move to MIA → confirm or enter qty → clear staging → insert staging row → commit_mass_transfer → beep → refresh."
Enter Barcode	"Flow: type barcode → duplicate check → lookup part, warehouse, category, part group → auto-select category and part group → find part in list → if missing load from barcode's warehouse and inject row → select part."
Check Up	"Flow: choose warehouse → list count dates with totals → choose date → list users with totals → choose user → list parts counted; Date Filter off shows all dates."
Close	"Flow: on close save window size/location/state locally → delete and insert app config rows for CYCLECOUNT → ignore errors."

```mermaid
flowchart TD
```

```
  S[Start app] --> L[Load batch: permissions, warehouses,<br/>storage locations, user units, App_Config]
  L --> SM[SecureMenu]
  SM --> WH[Select default warehouse] --> CAT[Load categories]
  CAT --> PG[Select category → load part groups]
  PG --> PL[Select part group → SP parts list]
  PL --> SEL[Select part]
  SEL --> DET[Details + dims + locations]
  DET --> MAC{Mark As Counted?}
  MAC -->|Confirm Yes| INS[(INSERT WarehouseCycleCounts)] --> PL
```

B6. Error Handling Rules + Retry Logic

Area	Handling	Retry
Mark As Counted	`Try/Catch` → message with `ex.Message`	None (user re-clicks)
Check-in (barcoded)	`SqlException` caught → generic fail path; typed `ErrorType` handled; form turns red	None
Check-in (non-barcoded)	`SqlException` caught; typed errors silent	None
Warehouse confirmation	`NullReferenceException` → assume OK	None
Window persistence	All exceptions swallowed silently	None
Clipboard	`ExternalException` → info message	User prompted "Please try again." (manual)

Grid data errors	`dgParts_DataError` / `dgDetails_DataError` suppress formatting errors	n/a
Everything else (MIA move, Enter Barcode, Part Info, Excel, LAME, launchers, Check Up, startup)	**No handling** → unhandled exception dialog / crash	None
Timeouts	Parts SP 600 s; all other commands use the `Web Page is Unavailable` default (30 s)	No automatic retries anywhere
Transactions	No client-side transactions. The MIA batch runs as one command batch without an explicit transaction (**Assumption:** the SP handles atomicity)	

B7. Tables + Entity Relationship Mapping (inferred from joins)

```mermaid
erDiagram
  inpart ||--o{ inpartsub : "partno"
  inpart ||--o{ WarehouseCycleCounts : "partno"
  inpart ||--o{ partlocationhistory : "partno"
  glentities ||--o{ inpartsub : "entityno = batchno"
  glentities ||--o{ partlocationhistory : "entityno = batchno"
  glentities }o--|| pjproptype : "proptype"
  glentities ||--o{ pjjobbudexp : "entityno"
  inpart ||--o{ pjjobbudexp : "partno"
```

```
  warehouses ||--o{ inpartsub : "warehouse_code = bld"
  warehouses ||--o{ WarehouseCycleCounts : "warehouse_code =
warehouse"
  pjtfrusr ||--|| mudbmsinfo : "user_name = username"
  pjtfrusr }o--|| inUnitOfWeight : "unitofweight"
  pjtfrusr }o--|| inUnitOfLength : "unitoflength"
  mumenus ||--o{ fn_user_menus : "packagecd, menuname, menuseqno"
```

Table	Key columns referenced
`inpart`	partno, partdesc, commmodity, length_uom, width_uom, depth_uom, projpercent (weight), dimsVerifiedDate/User, weightVerifiedDate/User
`inpartsub`	partno, unique_no (barcode), parentunique_no, serial_no, batchno, bld, dept, area, row, onhand, receipt
`glentities`	entityno, entitydesc, proptype, industry, agency
`pjproptype`	proptype, isutilized, isAvailable
`pjjobbudexp`	entityno, partno, todate
`partlocationhistory`	partno, batchno, bld, onhand, startdate, enddate
`WarehouseCycleCounts`	warehouse, partno, lastcount, username
`warehouses`	warehouse_code, warehouse_description, is_default
`pjtfrusr`	user_name, owner_entity, warehouse_entity, currency,

	id_language, unitofweight, unitoflength
`inUnitOfWeight` / `inUnitOfLength`	conversionShortname, conversionFactorFromPounds / conversionFactorFromInches
`inStorageLocations`	storagelocation
`App_Config` / `AppConfig` / `app_config`	cmndsel, tag, value
`SysConfig`	tag, value
`mumenus`	packagecd, menuname, menuseqno, appframe
`MassTransferStaging_v`	session_id, tfrQty, partno, unique_no, serial_no, from/to BatchNo, from/to Dept, from/to Area, from/to Bld, row, parentunique_no

B8. Third-Party / Integration Details

Integration	Mechanism	Trigger	Timeout / Retry	Failure behaviour
Finesse SQL Server	`System.Data.SqlClient`	All features	600 s parts SP; 30 s default otherwise; no retry	Mostly unhandled
Infragistics UltraWinGrid 11.1	WinForms control (licensed)	Locations grid	n/a	Licence/DLL needed at build and runtime
Microsoft Excel	COM Interop (embedded types)	Excel button	n/a	Unhandled if Excel is missing
Other Finesse	`StartFinesseProces`	Toolbar/menu	n/a	Defined in ClairUtilities

EXEs	`s` (ClairUtilities)			
Podio issue tracker	`Process.Start(URL)`	LAME! button	n/a	Unhandled if URL is null
Windows audio / speech	`My.Computer.Audio.Play`; `AnnounceFlaggedPart`	Scan / check-in	async background	n/a
Clipboard	`System.Windows.Forms.Clipboard`	Copy menus	n/a	Message on failure
ClairUtilities	Git submodule (compile-time)	Build	n/a	**Missing → build fails**

B9. Business Rules Table

BR	Rule	Source
BR01	The warehouse list excludes `LAX2` (main) and `NEW`, `LAX2` (Check Up)	`Utility.vb:220`, `CheckUp.vb:35`
BR02	The user's warehouse is the warehouse flagged `is_default = 1`	`:363`, `:56`
BR03	Categories/part groups only include parts with on-hand > 0 in the selected warehouse on projects whose property type is utilized	`:385-447`

	or available; commodity `UNUSED` / blank excluded	
BR04	Part group = text before the first comma in the part description; parts without a comma never appear in a named group	`:437`
BR05	Parts ranking and quantities come solely from `Get_Cycle_Count_Utility_Parts_List`	`:1995`
BR06	Utilization colour: 0 → red; < 20% pink; ≥ 80% green; otherwise blue	`:540-553`
BR07	A cycle count is a new history row (insert-only) stamped with the server time and SQL login	`:939-940`
BR08	Counts are recorded against the user's profile warehouse (`pjtfrusr.warehouse_entity`), not the warehouse being viewed	`:940`
BR09	Location history covers the last 12 months and only phases whose property type is available	`:584-598`

BR10	Part cycle-count history spans all warehouses	`:630-633`
BR11	Dimensions are stored in inches and weight in pounds (`projpercent`), converted to the user's units for display (2 decimals)	`:1121-1129`
BR12	Verification stamps use `dbo.today()` (date) and the SQL login	`:1471, 1489`
BR13	Location rows: past end date → red text; child (has parent barcode) → grey italic; scanned → green	`:1440-1455`
BR14	`INSTOCK` and `REPAIR` are special projects: a scan there confirms presence (receipt date refresh), it does not check in	`:124, 1871-1878`
BR15	Scanning a barcode on any other project offers check-in to the user's default warehouse `INSTOCK`	`:150-162`
BR16	Check-in always targets `to_batchno = INSTOCK` , `hold_in_warehouse = False` , `allow_warehouse_transfer = True`	`:1616-1623, 1714`

BR17	INSTOCK barcode warehouse is taken from `inpartsub.bld` because the INSTOCK agency is `LITZ`	`:1805`
BR18	The warehouse mismatch warning can be suppressed for the rest of the session ("Don't show again")	`:1787, 1861`
BR19	If check-in returns a parent project of the requested phase, it is treated as "checked out to another project" (warning)	`:1644-1650`
BR20	Barcoded items move qty 1; non-barcoded items move a user-entered qty ≤ on-hand	`:1355-1369`
BR21	The MIA transfer keeps dept/area/row, changes phase to the MIA order and warehouse to the MIA project's agency; only top-level (non-child) lines	`:1376-1402`
BR22	The MIA order is a per-user setting (DB `App_Config` overrides the local setting at startup)	`:240-241`
BR23	Menu rights: Utilization, Add Inventory, Device Maintenance (Remove	`:276-344`

	Inventory), Part Maintenance come from `fn_user_menus`; Add/Remove also need `can_edit_inventory`, Part Maintenance needs `can_edit_parts`	
BR24	Excel export excludes `username` but includes other hidden columns	`:828-829`
BR25	Enter Barcode accepts barcodes with on-hand > 0 or child barcodes; parts from another warehouse are injected with blank quantities and last count 1900-01-01	`:1937, 1965-1975`
BR26	Window layout is saved per user under `cmndsel = 'CYCLECOUNT'`	`:14, 79-83`

B10. User Access Matrix

Legend: ✔ allowed, ✖ hidden/blocked, — not applicable. Columns: **All** = any user with a `pjtfrusr` profile; **Util** = has the "Utilization" menu right; **Inv** = `can_edit_inventory` + the corresponding menu right; **Parts** = `can_edit_parts` + "Part Maintenance" right; **NoWH** = no default warehouse.

Function	All	Util	Inv	Parts	NoWH	Enforcement
Browse / filter /	✔	✔	✔	✔	⚠ (no default	none

search parts					selection)	
Part history views, dims status bar	✔	✔	✔	✔	✔	none
Verify dimensions / weight (writes `inpart`)	✔	✔	✔	✔	✔	**none**
Mark As Counted (writes)	✔	✔	✔	✔	✔	**none**
Cycle Count barcode scan / check-in	✔	✔	✔	✔	✔ (uses `is_default` query)	none
Check item into your warehouse	✔	✔	✔	✔	✖	client check `_UserWarehouseCode`
Move to MIA / Reset MIA	✔	✔	✔	✔	✔	none
Availability,	✔	✔	✔	✔	✔	none

Query Tools						
Utilization (toolbar)	✖	✔	✖	✖	—	menu right
Utilization (context menu)	✔	✔	✔	✔	✔	**none (bypass)**
Add Inventory	✖	✖	✔	✖	—	menu right + `can_edit_inventory`
Remove Inventory	✖	✖	✔	✖	—	menu right "Device Maintenance" + `can_edit_inventory`
Part Maintenance	✖	✖	✖	✔	—	menu right + `can_edit_parts`
Project Maintenance, Transfer Tools	✔	✔	✔	✔	✔	none (target apps may enforce their own security

						– **Assumption**)
Excel export, Copy, LAME!	✔	✔	✔	✔	✔	none
Cycle Count Check Up (all users' counts)	✔	✔	✔	✔	✔	**none**

Server-side security (SQL permissions on tables/SPs) is not visible in this repository and may further restrict writes (**Assumption**).

B11. Edge Cases

Data edge cases

#	Scenario	Current behaviour
E01	User has no `pjtfrusr` row or unit mapping	Startup works; F10 throws on `_dtUserInfo.Rows(0)`
E02	No default warehouse	Combo unselected; `CurrentWarehouseCode()` on empty text → `Left(…, -1)` error when categories load; check-in blocked
E03	Part deleted between load and select	F10 `Rows(0)` → exception
E04	Barcode `commmodity` NULL or description without comma (Enter Barcode)	`InvalidCastException`

E05	Scan barcode belonging to another part	Not found → fail beep, no message
E06	Scan → check-in fails	Row still turns green, local cells changed (false success)
E07	Check-in child barcode (`onhand` ≠ 1) via menu	`IndexOutOfRange` crash
E08	Non-barcoded qty = 0 or negative	Silently ignored
E09	MIA qty negative	Passes validation → negative transfer sent to the SP
E10	MIA qty decimal (e.g. 1.5)	`Integer.Parse` exception (unhandled)
E11	MIA move on child barcode	Inserts 0 rows; success beep anyway
E12	`SysConfig` URL missing	Crash on LAME!
E13	Project/Transfer Tools while details are in a Cycle Count History mode	`entityno` missing → exception
E14	Search text contains `[`, `]`, `*`	DataView filter error
E15	Check-in SP returns DBNull outputs	Cast error not caught
E16	Mark As Counted with `mudbmsinfo` not user-scoped	Multiple rows, one per user/warehouse

E17	Partial success: barcoded check-in error after the flag check	Flagged sound already played, then fail; form stays red
E18	Excel not installed	COM exception (unhandled)
E19	Parts list with more than 26 columns in Excel	Column-letter defect (range mis-sized)
E20	Window minimized at close	Restored as neither Normal nor Maximized (`Select Case` has no branch)

Timeout & retry

- Parts SP: 600 s; others use the Web Page is Unavailable default of 30 s. No retry or back-off anywhere; the user must repeat the action.
- The UI blocks during every query (no async).

Security validation present in code

- Windows identity → `SUSER_SNAME()` stamped on all writes.
- Menu-right trimming (4 items) and 2 DB permission functions.
- `SQLQuote()` used for most literals (implementation not visible). **Unquoted concatenations:** `Populate_PartHistory` (`partno`, warehouse), `Populate_PartCycleCountHistory` (`partno`); `Integer.Parse` protects the MIA quantity.
- No input length limits; no client-side audit log.

Performance constraints / bottlenecks

- Parts SP can run up to 10 minutes (timeout 600 s) on the UI thread.
- The filter timer can spin every 300 ms indefinitely (F08 gap).
- Category/part-group queries scan `inpart` × `inpartsub` × `glentities` × `pjproptype` with `CHARINDEX` / `CROSS APPLY` (not SARGable).
- Check Up runs the dates query twice on open.
- Excel export builds a DataTable cell by cell from the grid (fine for small lists).
- A long-lived shared connection per form; Check Up never disposes its connection.

B12. Gaps Summary (as-is defects / missing logic)

#	Gap
G01	`ClairUtilities` submodule missing – the repo cannot be built or fully specified
G02	"Search all part groups" fallback is ineffective (flag not passed to `Get_Parts_From_Warehouse`)
G03	Filter timer not stopped on an unchanged search
G04	Scan check-in ignores the check-in result (false green) and skips the warehouse confirmation
G05	Config persistence uses three different table names (`app_config`, `AppConfig`, `App_Config`)
G06	The MIA order chosen in the app is not saved to the DB; the DB value overrides it at next startup
G07	Write actions (Mark As Counted, verify, check-in, MIA) have no permission checks; context-menu Utilization bypasses the menu right
G08	MIA move lacks error handling, allows negative quantities, crashes on decimals, beeps success even when nothing moved
G09	Form background stays red after any check-in error
G10	Hidden "Export Part History to Excel" button with no implementation
G11	Check-in sound played twice on barcoded success

G12	Mark As Counted query has no explicit current-user filter (depends on `mudbmsinfo` scope)
G13	Unused files (`Index.dat` , `Storage.dat` , 4 icons) and dead code (section 0.3)